



G₂O Water Technology and Applications Overview

2015
Contact Tim Harper
+44 7894 708989
tim@g2o.co

G2O Water International Ltd

- Based in Manchester (UK) - the home of graphene
- Licensed technology from the University of South Carolina
- Board includes
 - Tim Harper - serial technology entrepreneur
 - Andrew Greenaway - former head of strategy at SIG Plc a FTSE 250 company
 - Jeremy Pelczer- former CEO Thames Water & American Water, Chair of WaterAid.

The Unmet Market Need

- The world is forecast to face a 40% global shortfall between water demand and available supply by 2030 (UN)
- Within the next 15-20 years, the worsening water security situation risks triggering a global food crisis, with shortfalls of up to 30% in cereal production (UN)
- Global membrane market forecast to be \$32.9bn by 2019
- G2O technology generates significant value in the form of major operating cost savings for the ultimate end user which due to the relatively low incremental cost of deployment also makes it an attractive proposition for the membrane manufacturer.

G₂O Coating Technology

- Small graphene oxide (GO) flakes are laid on a commercially available filter.
- The flakes wrap tightly around the polyamide fibres and act as a functional coating for modifying the surface properties of the polyamide.
- A pi-pi interaction between the flakes and fibres also ensures stability, even with the shear stresses resulting in cross-flow operation.

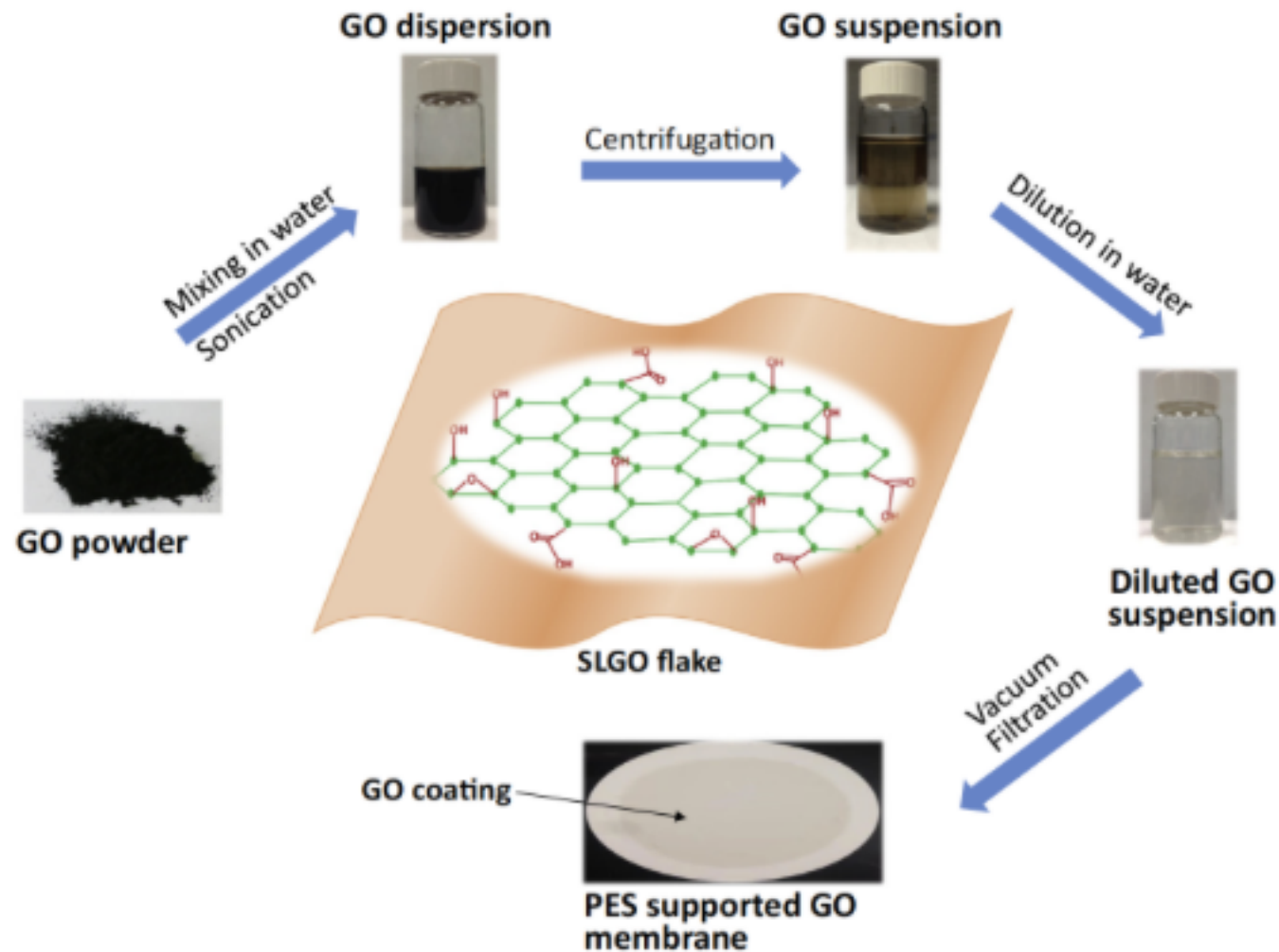


Graphene coated polyamide membrane

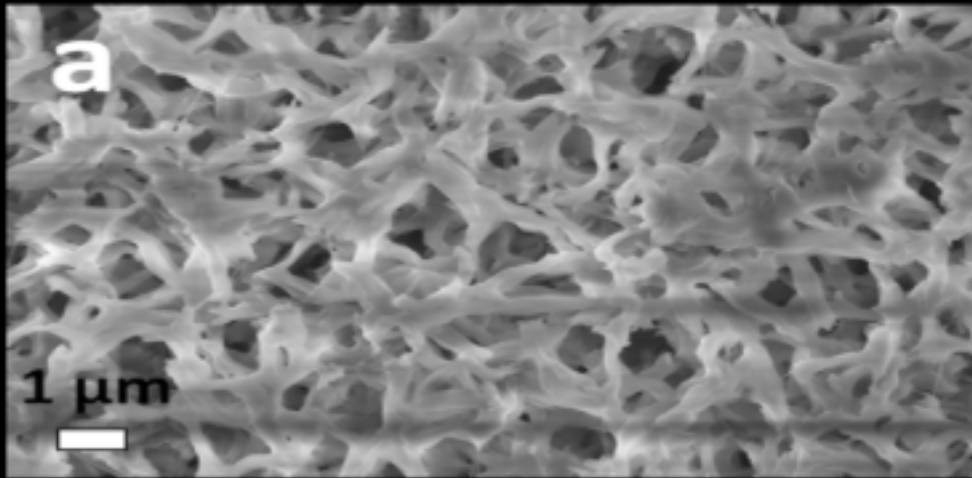
G₂O Graphene Technology

- G₂O's patented technology graphene oxide acts as a functional coating for modifying the surface properties of existing filter media resulting in
 - Higher pure water flux, High fouling resistance, Excellent mechanical strength, High chemical stability, Good thermal stability, Low cost
- By varying the graphene oxide concentration in water, G₂O' membranes with different thickness can be easily fabricated via a one-time filtration process.
- G₂O uses graphene membranes that gives a unique combination of high flux rates 100 to 1000 times greater than current ultrafiltration membranes.
- The nanochannels between graphene oxide sheets allow water to pass through, while rejecting larger contaminants, similar to the sieving mechanism of membranes.
- The membranes can recover from fouling without the need for chemicals and the pore sizes can be tuned to specific applications such as water desalination, oil/water separation, storm water treatment and industrial waste water recovery.

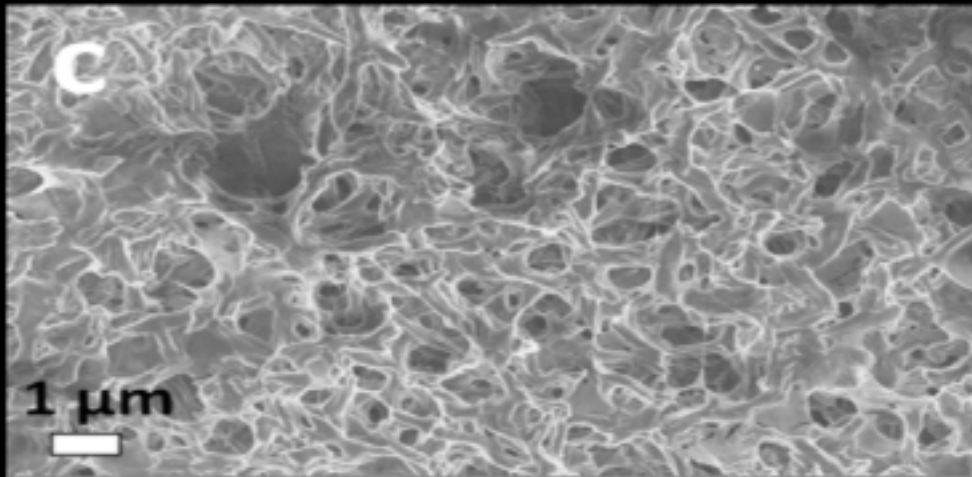
Membrane Production



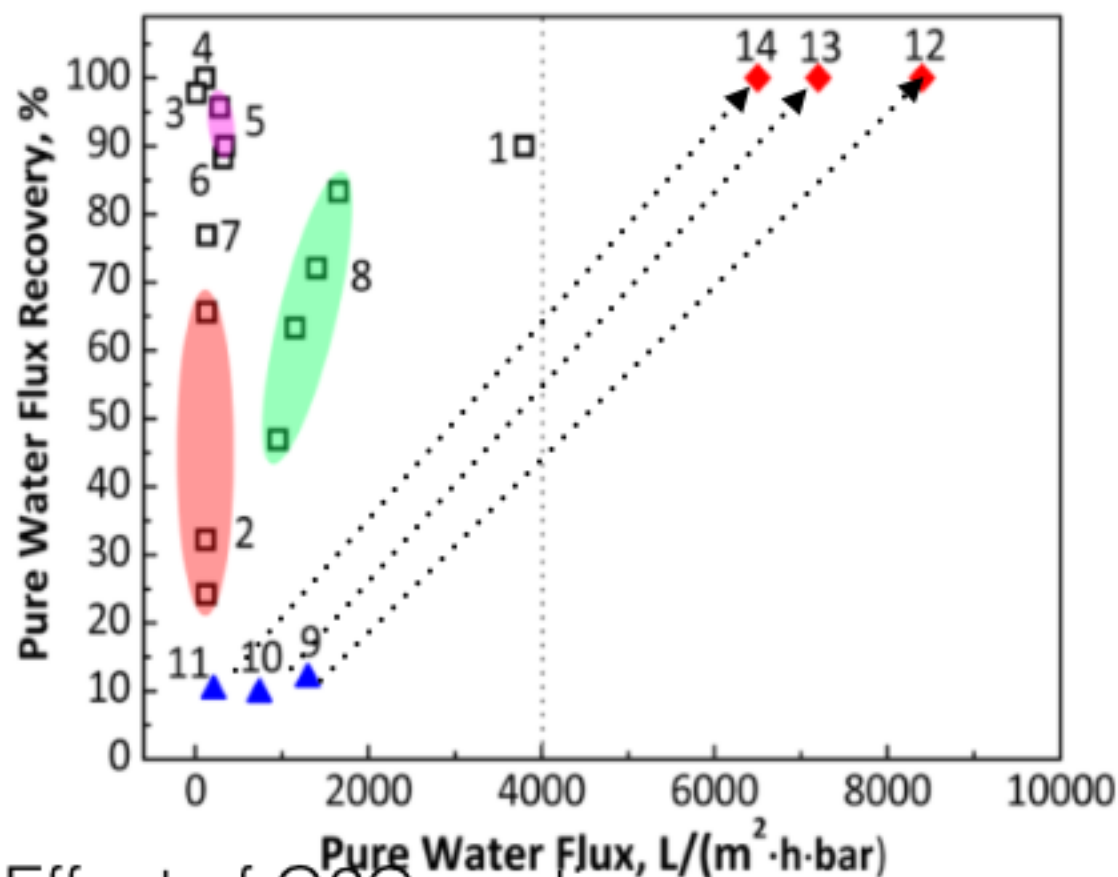
G₂O Coating Technology



200 nm pore Polyamide
membrane



Polyamide membrane
coated with 15nm of
graphene oxide

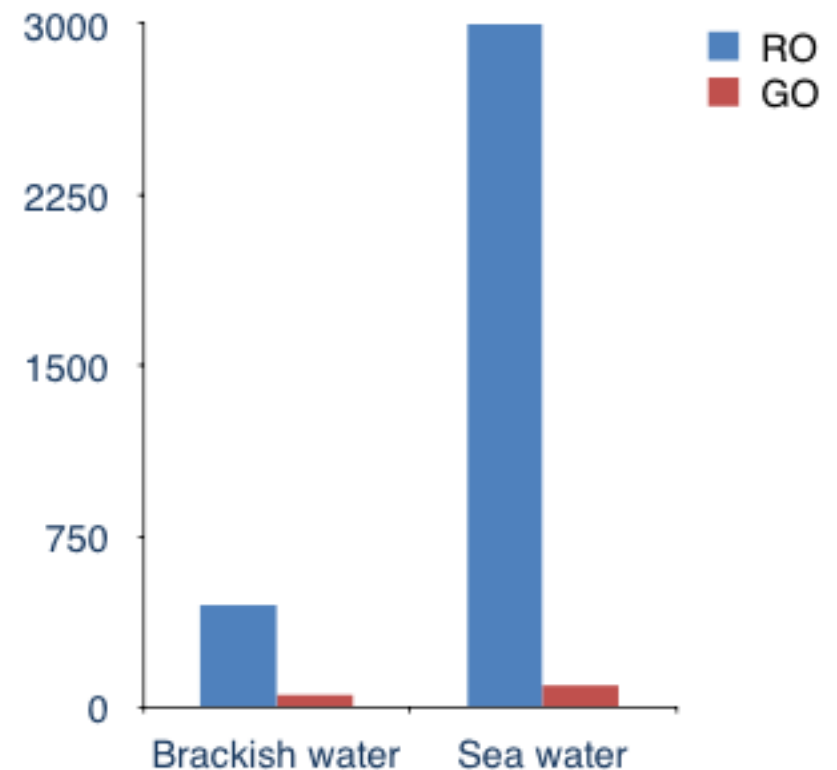


No	Symbo	Membrane	Reference
1	□	CaCO ₃ /PAA-g-PP	(9)
2	□	PES/Pluronic F127	(24)
3	□	cellulose/NMMO·H ₂ O/PE	(26)
4	□	PAN-g-PEO	(27)
5	□	CA-g-PAN	(28)
6	□	nano-sized ZrO ₂ /Al ₂ O ₃	(29)
7	□	ZrO ₂ /α-Al ₂ O ₃	(30)
8	□	EVAL/PEO-PPO-PEO	(31)
9	▲	PA	This work
10	▲	CN	
11	▲	AAO	
12	◆	10-nm GO on PA	
13	◆	15-nm GO on CN	
14	◆	15-nm GO on PA	

Effect of G2O coating on commercially available membranes showing dramatic increase in both flux (volume of water, horizontal axis) and recovery (% of water transmitted through the

Water Treatment

- Higher flux means lower operating pressures
- Graphene filter technology potential to reduce energy costs by up to 97%.
- Applications include
 - Desalination of seawater
 - Environmental maritime applications in aquaculture and oil & gas production
 - Drain water and waste water management



Energy (kW) required to desalinate 1 m³ of water with RO compared to GO membrane filtration

Water Treatment

Flux Comparisons
(higher numbers are better)

<i>Membrane</i>	<i>Manufacturer</i>	<i>Pure Water Flux (l/m²/hr/bar)</i>
SR3D	Koch	0.35
Filmtec NF270	Dow	13.3
DL	GE Osmonics	3.06
G ₂ O	G ₂ O	500 (8000 for ultrafiltration)

Maximum Energy Savings for Desalination*

<i>Brackish water</i>	<i>Seawater</i>
80%	97%

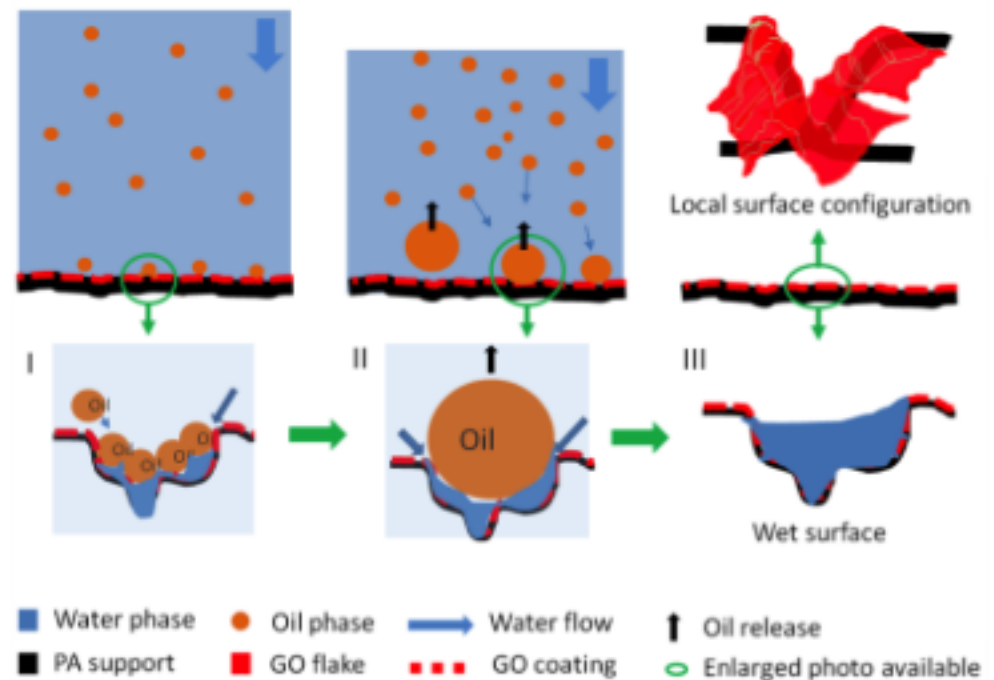
*Assumes 10 bar pressure required for brackish water, 70 bar for seawater using current RO membranes and 50% energy recovery

Oil/Water Separation

- Fouling of nano/ultrafiltration membranes in oil/water separation is a longstanding issue and a major economic barrier for their wide application. Currently reported membranes typically show severe fouling, resulting from the strong adhesion of oil on membrane surface and/or oil penetration inside the membranes. This greatly degrades their performance and shortens service lifetime. G₂O uses graphene oxide (GO) for the fabrication of fully-recoverable membranes for high flux, antifouling oil/water separation via function and structure mimicking of fish scales.
- The ultrathin, amphiphilic, water-locking GO coating mimics the thin mucus layer covering fish scales, while combination of corrugated GO flakes and intrinsic roughness of the porous supports successfully reproduces the hierarchical roughness of fish scales. Cyclic membrane performance evaluation tests reveal ~100% membrane recovery by facile surface water flushing, establishing their excellent easy-to-recover capability.
- Initial water flux can be as high as 4400 L/m².h.bar. The ultrathin GO membrane shows excellent superoleophobicity (168 degree contact angle) and anti-fouling property and a super high water flux. The ultrathin GO membrane can be easily recovered without loss of initial water flux.

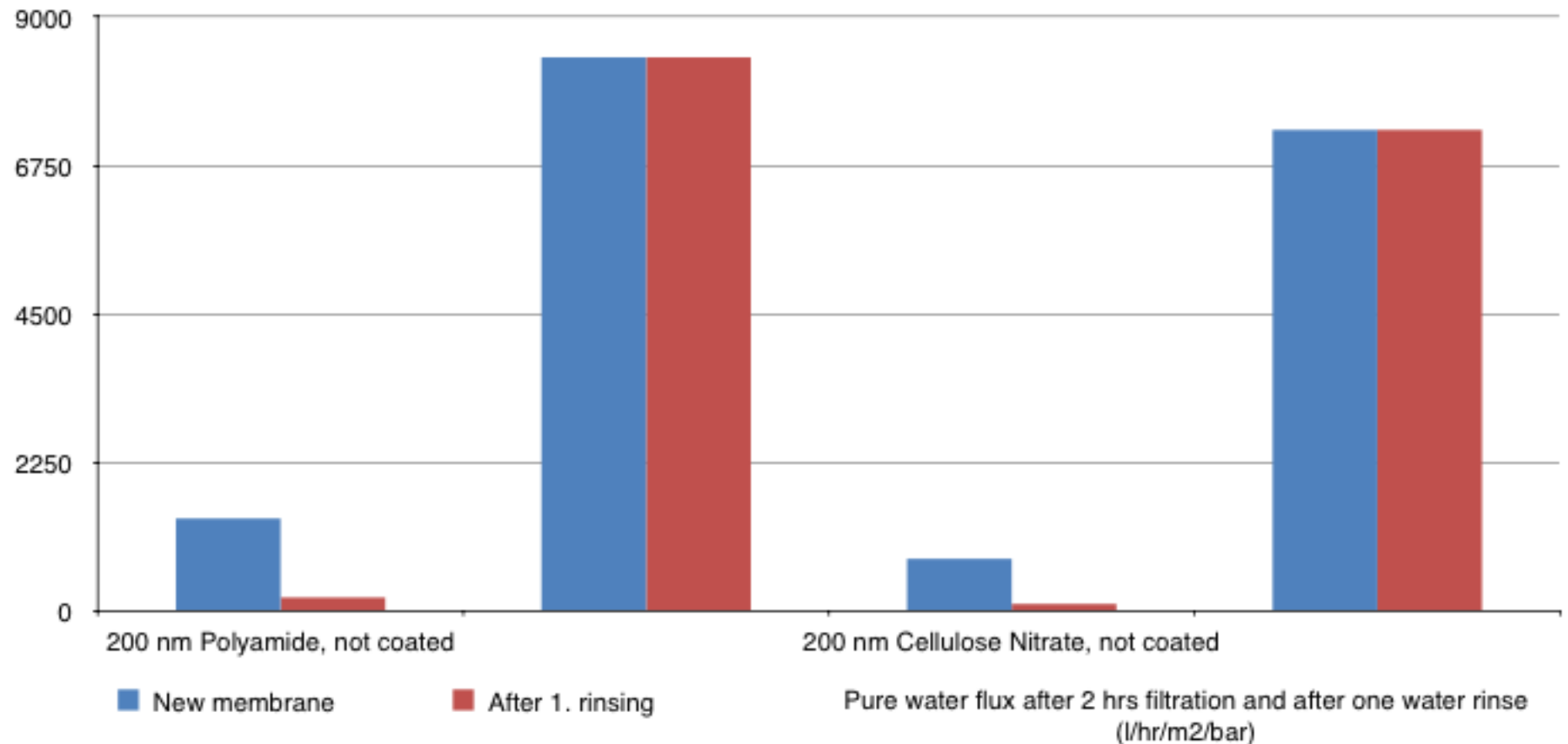
Oil/Water Separation

- In the G_2O oil/water separation process, water preferentially permeates through the GO coated membranes, forming numerous water pockets inside the structure, which serve as a repulsive support for oil droplets.
- The hydrophilic GO coating helps trap the water and reduces the effective permeation pore size, introducing additional nano-scaled roughness for smaller oil droplets repulsion.
- Oil droplets are repelled to the membrane surface and accumulate to form bigger droplets.
- Then, the oil is released from the super-oleophobic GO membrane surface and floats on top of the oil/water mixture.



Schematic illustration of underwater antifouling behaviour of GO coated PA surface in the process of oil/water emulsion separation. (I) oil droplets in contact with GO coated membrane surface, (II) accumulation of oil droplets and release of big oil droplets, and (III) GO coated PA membrane after water flushing/cleaning.

Fouling Is No Longer An Issue



Current Status

- Patent granted
- \$1m awarded by UK government to develop smart water filters to clean waste water. Partners include the National Nuclear Laboratory.
- Project via the World Economic Forum to develop a gravity fed water purification system for off grid communities.
- Raising a further \$5 million investment